

2. In each part, explain why the set S is not a spanning set for the vector space V .

(a) $S = \{t^3, t^2, t\}, V = P_3$

(b) $S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}, V = R^2$

(c) $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}, V = M_{22}$

2. (a) 1 does not belong to $\text{span } S$.

(b) $\text{Span } S$ consists of all vectors of the form $\begin{bmatrix} a \\ 0 \end{bmatrix}$, where a is any real number. Thus, the vector $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is not in $\text{span } S$.

(c) $\text{Span } S$ consists of all vectors of M_{22} of the form $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$, where a and b are any real numbers.

Thus, the vector $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is not in $\text{span } S$.

4. In each part, determine whether the given vector A in M_{22} belongs to $\text{span } \{A_1, A_2, A_3\}$, where

$$A_1 = \begin{bmatrix} 1 & -1 \\ 0 & 3 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix},$$

$$\text{and } A_3 = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}.$$

(a) $A = \begin{bmatrix} 5 & 1 \\ -1 & 9 \end{bmatrix}$ (b) $A = \begin{bmatrix} -3 & -1 \\ 3 & 2 \end{bmatrix}$

(c) $A = \begin{bmatrix} 3 & -2 \\ 3 & 2 \end{bmatrix}$ (d) $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

4. (a) Yes. (b) Yes. (c) No. (d) No.

12. Find a set of vectors spanning the null space of

$$A = \begin{bmatrix} 1 & 1 & 2 & -1 \\ 2 & 3 & 6 & -2 \\ -2 & 1 & 2 & 2 \\ 0 & -2 & -4 & 0 \end{bmatrix}.$$

12. $\left\{ \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$

14. The set W of all 3×3 matrices A with trace equal to zero is a subspace of M_{33} . (See Exercise 11 in Section 4.3.) Determine a subset S of W so that $\text{span } S = W$.

$$14. S = \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \right\}.$$

1. Show that

$$S = \left\{ \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 10 \\ 0 \\ 10 \end{bmatrix} \right\}$$

is a linearly dependent set in R^3 .

2. Show that

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

is a linearly independent set in R^3 .

Solution.

1. We form Equation (1):

$$c_1 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} 10 \\ 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

which has nontrivial solutions. Hence, S is linearly dependent.

2. We form Equation (1):

$$c_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

which has only the trivial solution. Hence, S is linearly independent.

2. Which of the following sets of vectors are bases for R^3 ?

- (a) $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$
- (b) $\left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$
- (c) $\left\{ \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$
- (d) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

solution: 2. (c).

4. Which of the following sets of vectors are bases for P_2 ?

- (a) $\{-t^2 + t + 2, 2t^2 + 2t + 3, 4t^2 - 1\}$
- (b) $\{t^2 + 2t - 1, 2t^2 + 3t - 2\}$
- (c) $\{t^2 + 1, 3t^2 + 2t + 1, 6t^2 + 6t + 3\}$
- (d) $\{3t^2 + 2t + 1, t^2 + t + 1, t^2 + 1\}$

4. (d).

6. Show that the set of matrices

$$\left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \right\}$$

forms a basis for the vector space M_{22} .

6. If $c_1 \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, then $\begin{bmatrix} c_1 + c_3 & c_1 + c_4 \\ c_2 + c_4 & c_2 + c_3 + c_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. The first three entries imply $c_3 = -c_1 = c_4 = -c_2$. The fourth entry gives $c_2 - c_2 - c_2 = -c_2 = 0$. Thus $c_i = 0$ for $i = 1, 2, 3, 4$. Hence the set of four matrices is linearly independent. By Theorem 4.12, it is a basis.

In Exercises 7 and 8, determine which of the given subsets forms a basis for R^3 . Express the vector

$$\begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

as a linear combination of the vectors in each subset that is a basis.

$$8. \text{ (a) } \left\{ \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 5 \\ 1 \end{bmatrix} \right\}$$

$$\text{(b) } \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ -1 \end{bmatrix} \right\}$$

$$8. \text{ (b) is a basis for } R^3 \text{ and } \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} - 1 \begin{bmatrix} 3 \\ 4 \\ -1 \end{bmatrix}.$$

In Exercises 9 and 10, determine which of the given subsets form a basis for P_2 . Express $5t^2 - 3t + 8$ as a linear combination of the vectors in each subset that is a basis.

$$9. \text{ (a) } \{t^2 + t, t - 1, t + 1\} \quad \text{(b) } \{t^2 + 1, t - 1\}$$

$$10. \text{ (a) } \{t^2 + t, t^2, t^2 + 1\} \quad \text{(b) } \{t^2 + 1, t^2 - t + 1\}$$

$$10. \text{ (a) forms a basis: } 5t^2 - 3t + 8 = -3(t^2 + t) + 0t^2 + 8(t^2 + 1).$$

12. Find a basis for the subspace W of R_4 spanned by the set of vectors

$$\left\{ \begin{bmatrix} 1 & 1 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 1 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & -6 & -3 \end{bmatrix}, \begin{bmatrix} -1 & -5 & 1 & 0 \end{bmatrix} \right\}.$$

What is $\dim W$?

$$12. \text{ A possible answer is } \left\{ \begin{bmatrix} 1 & 1 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 3 & 1 \end{bmatrix} \right\}; \dim W = 3.$$

14. Let

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \right\}.$$

Find a basis for the subspace $W = \text{span } S$ of M_{22} .

$$14. \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}.$$

16. Find a basis for the subspace W of M_{33} consisting of all symmetric matrices.

$$16. \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}.$$

In Exercises 19 and 20, find a basis for the given subspaces of $\mathbb{R}^3$ and $\mathbb{R}^4$.

19. (a) All vectors of the form $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, where $b = a + c$

(b) All vectors of the form $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, where $b = a$

(c) All vectors of the form $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, where $2a + b - c = 0$

20. (a) All vectors of the form $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, where $a = 0$

(b) All vectors of the form $\begin{bmatrix} a + c \\ a - b \\ b + c \\ -a + b \end{bmatrix}$

20. (a) $\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$, (b) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \\ -1 \end{bmatrix} \right\}$,

22. Find a basis for the subspace of P_3 consisting of all vectors of the form $at^3 + bt^2 + ct + d$, where $c = a - 2d$ and $b = 5a + 3d$.

22. $\{t^3 + 5t^2 + t, 3t^2 - 2t + 1\}$.

In Exercises 23 and 24, find the dimensions of the given subspaces of $\mathbb{R}_4$.

24. (a) 3. (b) 2.

28. Find a basis for R^3 that includes

(a) the vector $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$;

(b) the vectors $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$.

28. (a) A possible answer is $\left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$. (b) A possible answer is $\left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}$.

30. Find a basis for M_{23} . What is the dimension of M_{23} ?
Generalize to M_{mn} .

30. $\left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\};$
 $\dim M_{23} = 6$. $\dim M_{mn} = mn$.

32. Find the dimension of the subspace of P_3 consisting of all vectors of the form $at^3 + bt^2 + ct + d$, where $b = 3a - 5d$ and $c = d + 4a$.

32. 2.

34. Give an example of a two-dimensional subspace of P_3 .

34. The set of all polynomials of the form $at^3 + bt^2 + (b - a)$, where a and b are any real numbers.

2. Let

$$A = \begin{bmatrix} 1 & 1 & -2 \\ -2 & -2 & 4 \\ -1 & -1 & 2 \end{bmatrix}.$$

(a) Find the set of all solutions to $A\mathbf{x} = \mathbf{0}$.

(b) Express each solution as a linear combination of two vectors in R^3 .

(a) $x = -r + 2s$, $y = r$, $z = s$, where r , s are any real numbers.

(b) Let $\mathbf{x}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$, $\mathbf{x}_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$. Then

$$\begin{bmatrix} -r + 2s \\ r \\ s \end{bmatrix} = r \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = r\mathbf{x}_1 + s\mathbf{x}_2.$$

In Exercises 3 through 10, find a basis for and the dimension of the solution space of the given homogeneous system.

$$4. \begin{bmatrix} 1 & -1 & 1 & -2 & 1 \\ 3 & -3 & 2 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$4. \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 6 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}; \text{ dimension} = 3.$$

$$6. \begin{aligned} x_1 - x_2 + 2x_3 + 3x_4 + 4x_5 &= 0 \\ -x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 &= 0 \\ x_1 - x_2 + 3x_3 + 5x_4 + 6x_5 &= 0 \\ 3x_1 - 4x_2 + x_3 + 2x_4 + 3x_5 &= 0 \end{aligned}$$

$$6. \left\{ \begin{bmatrix} 4 \\ 3 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \\ 0 \\ 1 \end{bmatrix} \right\}; \text{ dimension} = 2.$$

$$8. \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

8. No basis; dimension = 0.

$$10. \begin{bmatrix} 1 & 2 & -3 & -2 & 1 & 3 \\ 1 & 2 & -4 & 3 & 3 & 4 \\ -2 & -4 & 6 & 4 & -3 & 2 \\ 0 & 0 & -1 & 5 & 1 & 9 \\ 1 & 2 & -3 & -2 & 0 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$10. \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 17 \\ 0 \\ 5 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}; \text{dimension} = 2.$$

In Exercises 13 through 16, find a basis for the solution space of the homogeneous system $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$ for the given scalar λ and given matrix A .

$$13. \lambda = 1, A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$$

$$14. \lambda = -3, A = \begin{bmatrix} -4 & -3 \\ 2 & 3 \end{bmatrix}$$

$$14. \left\{ \begin{bmatrix} -3 \\ 1 \end{bmatrix} \right\}.$$

$$16. \lambda = 3, A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

16. No basis.

In Exercises 17 through 20, find all real numbers λ such that the homogeneous system $(\lambda I_n - A)\mathbf{x} = \mathbf{0}$ has a nontrivial solution.

17. $A = \begin{bmatrix} 2 & 3 \\ 2 & -3 \end{bmatrix}$ 18. $A = \begin{bmatrix} 3 & 0 \\ 2 & -2 \end{bmatrix}$

19. $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 0 \end{bmatrix}$

20. $A = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & -3 \\ 0 & 4 & 5 \end{bmatrix}$

18. $\lambda = 3, -2$.

20. $\lambda = 1, 2, -2$

In Exercises 21 and 22, determine the solution to the linear system $A\mathbf{x} = \mathbf{b}$ and write it in the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$.

21. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$

22. $A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 1 & 2 \\ 1 & -3 & 2 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$

22. $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$, where $\mathbf{x}_p = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$, $\mathbf{x}_h = r \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$, r any number.